

or CO₂ levels. In this project, we can collect sensor data using a Raspberry Pi or simulate the inputs, send it over to InfluxDB (a time series database), and visualise all of it in Grafana. InfluxDB can be used to set things up, configure measurement tags, and send sensor data using either Python or the HTTP API. After getting something set up in InfluxDB, some of the cool visualisations in Grafana can be used to create dashboards that visualise real-time data with live updating graphs, colour thresholds, and alert notifications. This practical setup can have a wide range of applications — from smart home dashboards and environmental monitoring to industrial IoT applications.

From CSV to storyboard: A notebook-first approach

Believe it or not, some of the most interesting stories come out of spreadsheets. In this project, we can begin with a raw CSV file (could be from the government or customer feedback) and go through each phase: loading, cleaning, exploring, visualising, and finally, documenting our findings. The goal is to take our notebook and create a short story (report) that combines code, visuals, and clear language—a powerful tool for internal presentations or your blog.

Learning path: Becoming a data visualisation pro

Want to delve into creating jaw-dropping plots, dashboards, and data stories that really connect with people? Begin with basic plotting. Become comfortable plotting bar charts, line graphs, or pie charts in Excel, seaborn, or ggplot2. Get accustomed to the principles of scales, colour theory, and where each chart fits best. While fully grasping interactivity and storytelling may be still a bit beyond you, tools like Plotly, Altair, or Streamlit should be where you practice how raw data insights can turn into

Top open source libraries for plotting in Python, R, and JavaScript

Language	Library	Best for
Python	Matplotlib	Static, detailed scientific plots
	Seaborn	Statistical charts and aesthetics
	Plotly	Interactive dashboards and notebooks
	Altair	Declarative, concise syntax
R	ggplot2	Grammar of graphics, elegant plots
	Plotly R API	Interactive visuals for Shiny apps
	Lattice	Multivariate plots, conditioning
JavaScript	D3.js	Web-first, highly customisable visuals
	Chart.js	Quick charts for web dashboards
	Vega-Lite	Simpler, declarative syntax for the web

Troubleshooting common visualisation pitfalls

Pitfall	Fix
Cluttered visuals	Remove gridlines, reduce chart ink
Too many colours or labels	Stick to a colour palette and label key items
Misleading axis scales	Use zero-baselines for bar/line charts
Inconsistent units or timeframes	Normalise data before comparing
Ambiguous legends or titles	Use descriptive labels and chart titles

Free public datasets to practice your skills

Source	What you'll find
Kaggle datasets	Crowdsourced, beginner to advanced datasets
data.gov.in	Indian government datasets, health, education
UCI ML repository	Classic academic datasets for ML and EDA
FiveThirtyEight	Journalism-ready CSVs with context
Google Dataset Search	Searchable index of open data

engaging visual experiences. Build a dashboard that lets people explore trends through filters, sliders, or search boxes. For the real deal, go real-time with Dash, Grafana, or D3.js and learn chart animations, handling of user input, and connecting to live databases

or APIs. Of course, document your projects! Share on GitHub, publish via Streamlit Cloud, or embed within your blog posts. Nothing would impress your future employers or clients more than well-documented, interactive showcases of your work. **END** 

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